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KOTA (RAJASTHAN)
path to success

JEE (MAIN+ADVANCED)

Organic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Reduction

English Medium

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REDUCTION

REDUCING AGENTS AND THEIR ROLE

Group	Product	LAH in Ether	$NaBH_4$ in H_2O	B_2H_6 in THF	H_2 / Catalyst Δ
–CHO	$-CH_2OH$	+	+	+	+
>C = O	>CH – OH	+	+	+	+
$-CO_2H$	$-CH_2OH$	+	–	+	–
$-CO_2R$	$-CH_2OH$	+	–	+	+
–COCl	$-CH_2OH$	+	+	–	+
$-CONH_2$	$-CH_2NH_2$	+	–	+	+
$(RCO)_2O$	RCH_2OH	+	–	+	+
–CN	$-CH_2NH_2$	+	–	+	+
>C = NOH	$-CH_2NH_2$	+	–	–	+
>C = C<	>CH – CH<	–	–	+	+
–C ≡ C–	–CH = CH–	–	–	+	+
1° RX	RH	+	–	–	+

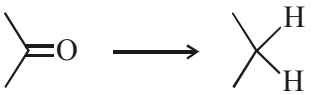
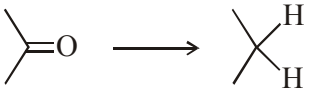
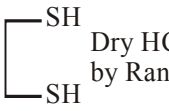
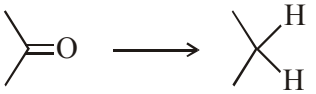
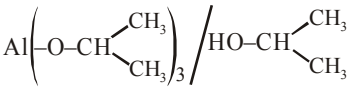
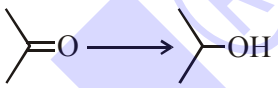
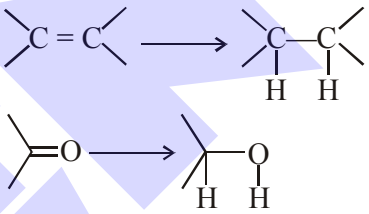
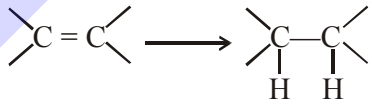
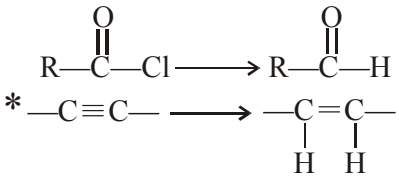
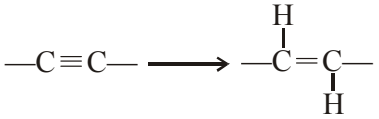
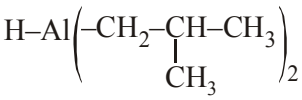
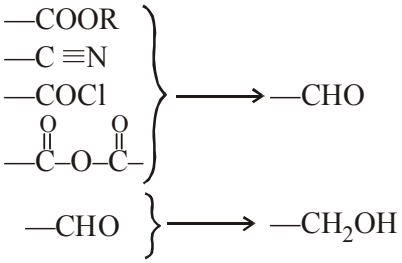
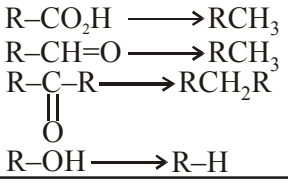
Note :

1. B_2H_6 in THF not able to reduce cyclic ester.

2. $NaBH_4$ can also reduce imine group ($\diagdown\!\!\diagup C = NH$).

3. Reactivity order towards Ni / H_2 , Δ ⇒ $-C \equiv N > -\underset{\underset{O}{\|}}{C} - O - R$

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Name	Reagent	Function
Wolf Kishner Reduction	(i) N_2H_4 / (ii) KOH, Δ	
Clemenson Reduction	Zn–Hg / HCl	
Mozingo Reduction	 Dry HCl, followed by Raney Ni	
Stephen's Reduction	$SnCl_2$ / HCl followed by H_3O^+	$R-C\equiv N \longrightarrow R-CH=O$
MPV Reduction		
Hydroboration Reduction	B_2H_6 / AcOH, H_2O	
Bouvoult Blank Reduction	Na / EtOH	$R-COO-R \longrightarrow RCH_2OH + ROH$
Transfer Hydrogenation	N_2H_4 / H_2O_2	
Rosenmund Reduction	H_2, Pd–$BaSO_4$	
Birch Reduction	Na / Liq. NH_3	
DIBAL-H (–78°C)	 followed by $H_3O^{\oplus}$	
Red phosphorus in presence of HI	Red P + HI	

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